Final presentation

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Product Marketing Engineering Intern | LPCS 2.4GHz

Project Overview

Objective	Develop developer-facing resources for low-power wireless connectivity
Technical Scope	BLE • BLE Mesh • Zephyr RTOS • OTA • CC2340R53 • CC2755X10
Starting Point	No prior wireless connectivity experience High-level project goals with limited guidance

Video 1 | Introduction to BLE Mesh

Technical Focus:

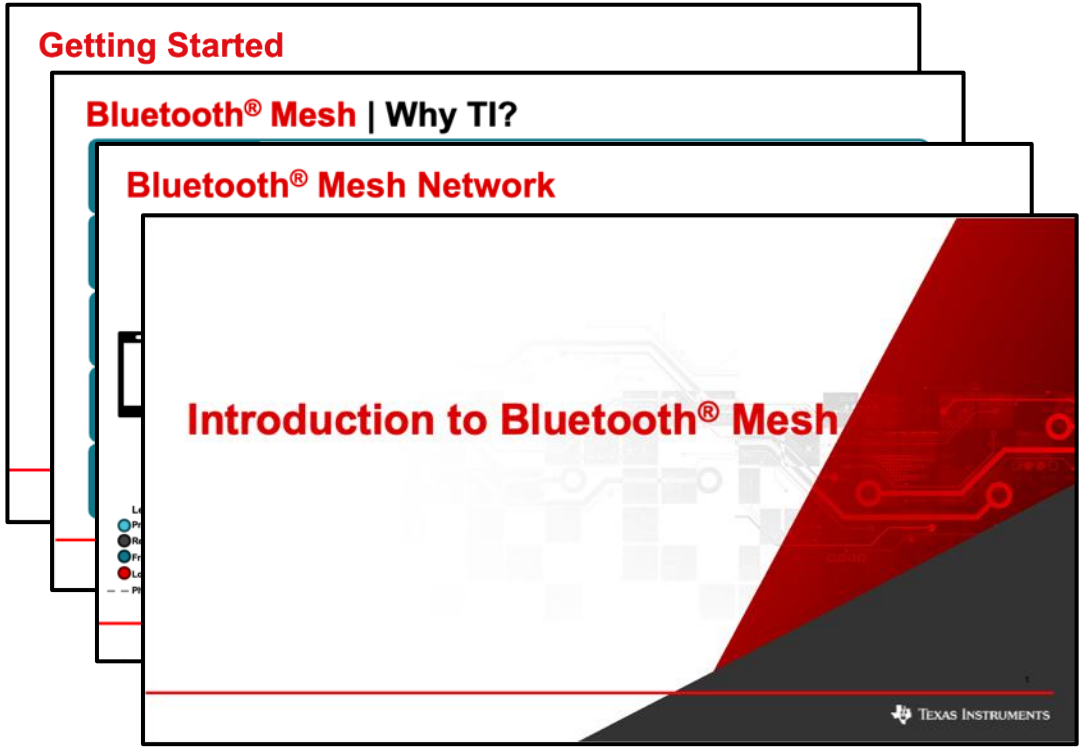
- BLE Mesh architecture
- Node functionality
- Scalability & security
- TI BLE Mesh Offering

Development:

- Started with no BLE Mesh knowledge
- Defined technical scope independently
- ~8 rounds of technical review

Result:

- Completed video
- Submitted through JIRA for TI.com publication



BLE Mesh | Technical Overview

Mesh Architecture

- Many-to-many communication
- Messages can be relayed across nodes
- Designed for large device networks

Node Functionality

- **Relay:** Retransmits mesh messages
- **Proxy:** Connects BLE devices to the mesh
- **Friend:** Stores messages for low-power nodes
- **Low Power Node (LPN):** Reduces radio activity to conserve energy

Video 2 | Building & Running a Zephyr BLE Demo

Technical Focus:

- Zephyr RTOS
- Zephyr BLE Peripheral example
- BLE characteristics
- GPIO control
- CC2340R53

Development:

- Modified existing Zephyr example
- Implemented BLE-controlled LED functionality
- Built, flashed & tested on hardware
- Troubleshot Linux/OpenOCD issues

Result:

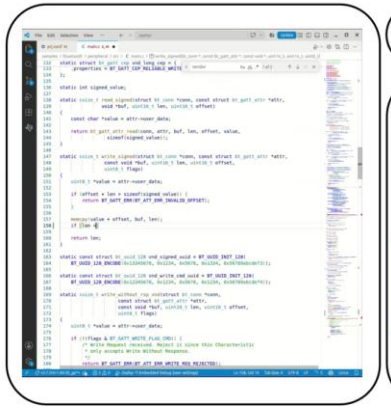
- Functional BLE demonstration
- Completed technical video
- Submitted through JIRA for TI.com publication

- **Demo**
  - Open TI Connect App on your phone
  - Filter by device name “Zephyr Bluetooth Demo”
  - Navigate to the vendor characteristic(second to bottom) and write a value

03:55.00







01:29.00

- **Change 3 – Toggle on Write**
  - Cmd+F – “vendor” & scroll to the second to last service
  - Right-click on write_signed & go to definition
  - You should be around line 147, find memcpy(value….);
  - Under that, add:
    If (len > 0) {
    gpio_pin_toggle_dt(&led); }

Zephyr BLE Demo | Technical Overview

BLE Implementation

- Zephyr BLE Peripheral example
- Writable BLE characteristic
- Characteristic write triggers LED control

Hardware Interaction

- CC2340R53 onboard LED
- GPIO configured through Zephyr
- LED controlled wirelessly through SimpleLink Connect

Video 3 | Zephyr BLE OTA Demo

Technical Focus:

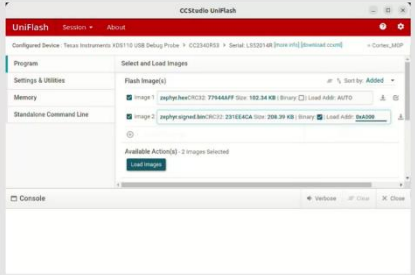
- Zephyr BLE OTA
- Firmware image updates
- SMP firmware transfer
- SimpleLink Connect
- CC2340R53

Development:

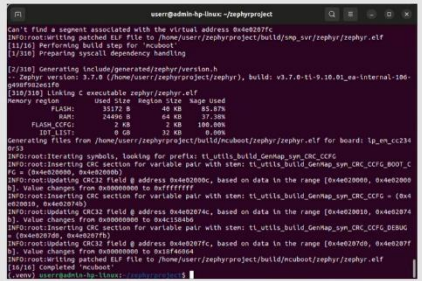
- Created distinct firmware versions
- Tested wireless firmware updates
- Troubleshot intended mobile OTA workflow

Result:

- Functional firmware completed
- OTA validated through alternative SMP workflow
- Intended mobile OTA workflow remained unreliable
- Final video not submitted for publication



- **Flash with UniFlash**
  - MCUBoot – build/mcuboot/zephyr/zephyr.hex
  - SMP Server – build/smp_svr/zephyr/zephyr.signed.bin
  - Change Load Addr to 0xA000
- **Connect via nRF Connect**
  - Open the nRF Connect App
  - Scan and connect to the LP



- **Build Firmware v2**
  - Same command for build, change v1 to v2
  - Transfer new zephyr.signed.bin to phone
- **OTA Update**
  - Navigate to the Firmware Update(SMP) in the top right
  - Select zephyr.signed.bin from your phone files
  - Upload and wait for the device to automatically reboot

Zephyr BLE OTA Demo | Technical Overview

OTA Process

- Firmware image generated from updated application
- Firmware transferred wirelessly using SMP
- Device reboots into updated firmware

Firmware Validation

- Created distinguishable firmware versions
- LED behavior verifies updated firmware
- Alternative SMP workflow validated OTA functionality

Side Project | Sales Simulation

Objective

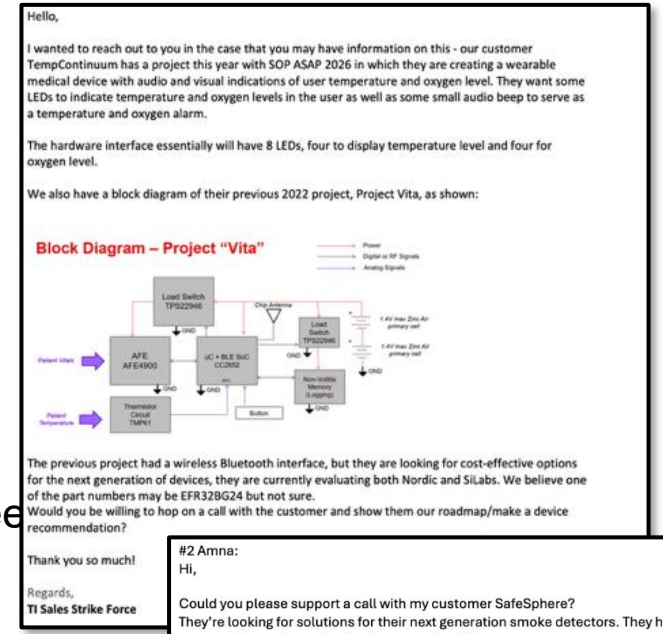
- Simulate a customer sales call with my team
- Recommend a TI solution based on customer requirements

Development

- Researched TI device roadmap and solutions
- Identified customer technical requirements
- Connected technical capabilities to customer nee

Result

- Completed two sales simulations
- Improved technical and customer communication



Hello,

I wanted to reach out to you in the case that you may have information on this - our customer TempContinuum has a project this year with SOP ASAP 2026 in which they are creating a wearable medical device with audio and visual indications of user temperature and oxygen level. They want some LEDs to indicate temperature and oxygen levels in the user as well as some small audio beep to serve as a temperature and oxygen alarm.

The hardware interface essentially will have 8 LEDs, four to display temperature level and four for oxygen level.

We also have a block diagram of their previous 2022 project, Project Vita, as shown:

Block Diagram – Project "Vita"

The previous project had a wireless Bluetooth interface, but they are looking for cost-effective options for the next generation of devices, they are currently evaluating both Nordic and SiLabs. We believe one of the part numbers may be EFR32BG24 but not sure.
Would you be willing to hop on a call with the customer and show them our roadmap/make a device recommendation?

Thank you so much!

Regards,
TI Sales Strike Force

#2 Amna:
Hi,

Could you please support a call with my customer SafeSphere?
They're looking for solutions for their next generation smoke detectors. They haven't decided which protocol they will go with. They're currently a SiLabs house.
I think they're currently on EFR32BG22 or BG24. I'm not entirely sure. They're not very open to share their project information, and it took us a lot of effort to get this meeting secured.
Need your help to get all the information you need from the call and push TI solution.

Best,
Field

Overall Results

Technical Deliverables

- 2 technical videos completed and submitted for TI.com publication
- OTA firmware and technical content completed
- Functional BLE and OTA demonstrations developed

Technical Development

- Progressed from no wireless experience to BLE, BLE Mesh, Zephyr & OTA
- Applied embedded programming through hands-on development
- Troubleshot hardware, software & development-environment issues

Professional Development

- Completed two customer sales simulations
- Collaborated across engineering and business functions
- Improved technical communication and independent problem solving

Key Takeaways





Technical Growth

- Built a foundation in BLE, BLE Mesh, Zephyr & OTA
- Gained hands-on embedded development experience
- Improved technical troubleshooting skills

Engineering Growth

- Learned when and how to ask focused technical questions
- Evaluated feedback instead of automatically implementing it
- Became more confident working independently

Career Takeaways

- Gained exposure to engineering and business functions
- Strengthened technical-to-customer communication
- Developed a stronger interest in technical engineering roles

Acknowledgements

- **Supervisor:** Xingyu Zou
- **Buddy:** Achyut Ray
- LPCS 2.4 GHz Connectivity Team
- Connectivity Interns









Questions?